Lecture 1(B): Quick introduction to flexibility

Dr. Md Umar Hashmi

KU Leuven & EnergyVille, Belgium

Agenda

- DERs in power networks
- Quick introduction to power system flexibility
- Flexibility parameters

Is growth of DER a good thing?

Yes

0%

No

0%

Don't Know

0%

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Do you know how flexibility is derived?

Yes

0%

No

0%

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DERs in power networks

What are distributed energy resources?

- **What are DERs?**
  - Small-scale, modular energy assets located close to where energy is used.
  - Capable of generating, storing, or managing electricity.
- **Common Examples**
  - **Photovoltaic (PV) Systems:** Rooftop solar panels.
  - **Electric Vehicles (EVs) & Charge Points:** Mobile storage and load.
  - **Batteries:** Stationary energy storage systems.
  - **Heat Pumps:** Efficient electric heating/cooling.
  - …

Source: https://www.gridx.ai/knowledge/distributed-energy-resources

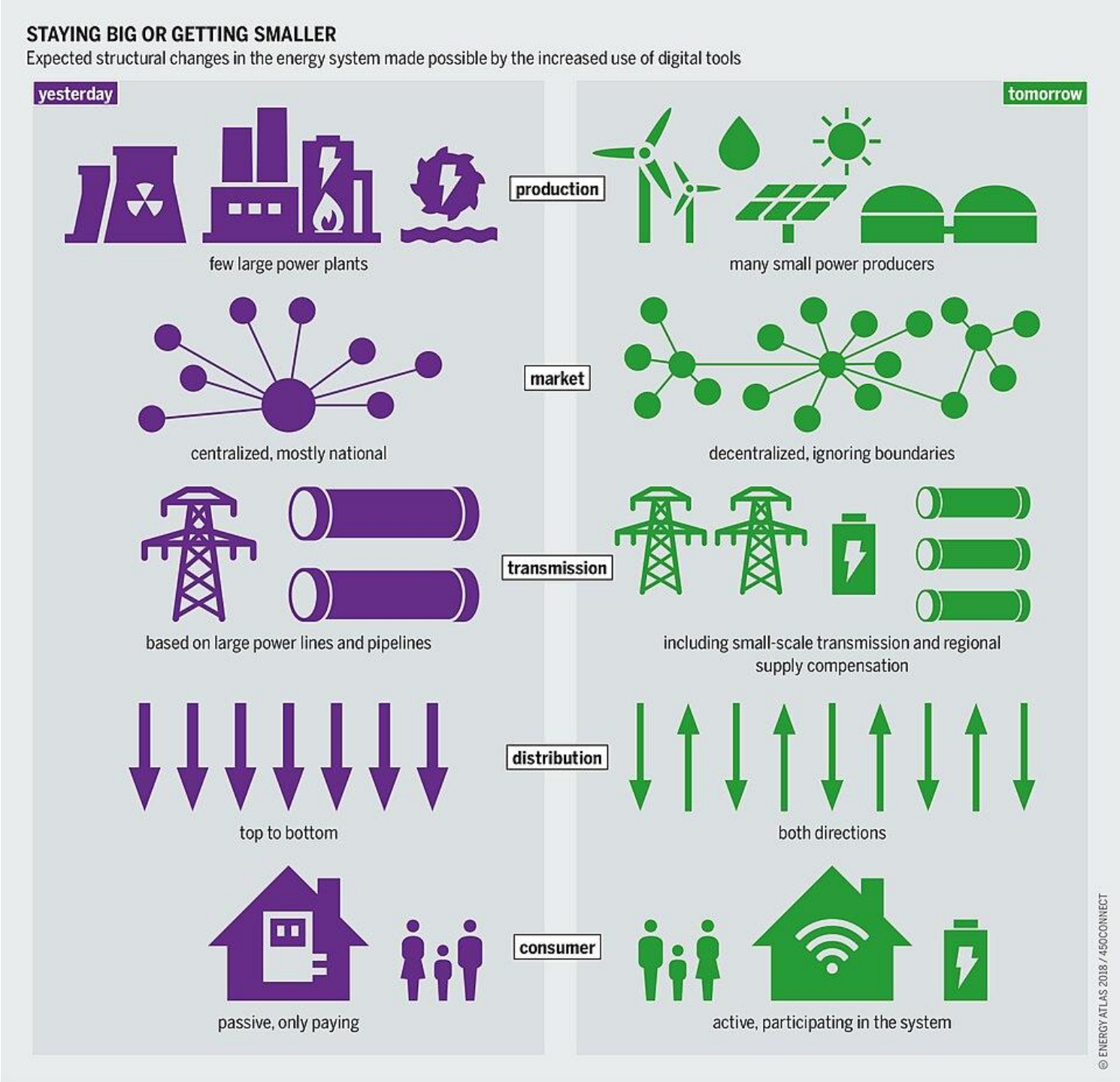
Challenges & opportunity with DER

- **The Challenge**
  - High penetration of unmanaged DERs can create imbalances between energy supply and demand.

- **The Opportunity (When Managed)**
  - **Improve Grid Reliability:** Intelligently integrating DERs enhances power system resilience.
  - **Increase Efficiency:** Reduces transmission losses by generating power locally.
  - **Support Decarbonization:** Lowers overall carbon emissions by enabling clean energy use.

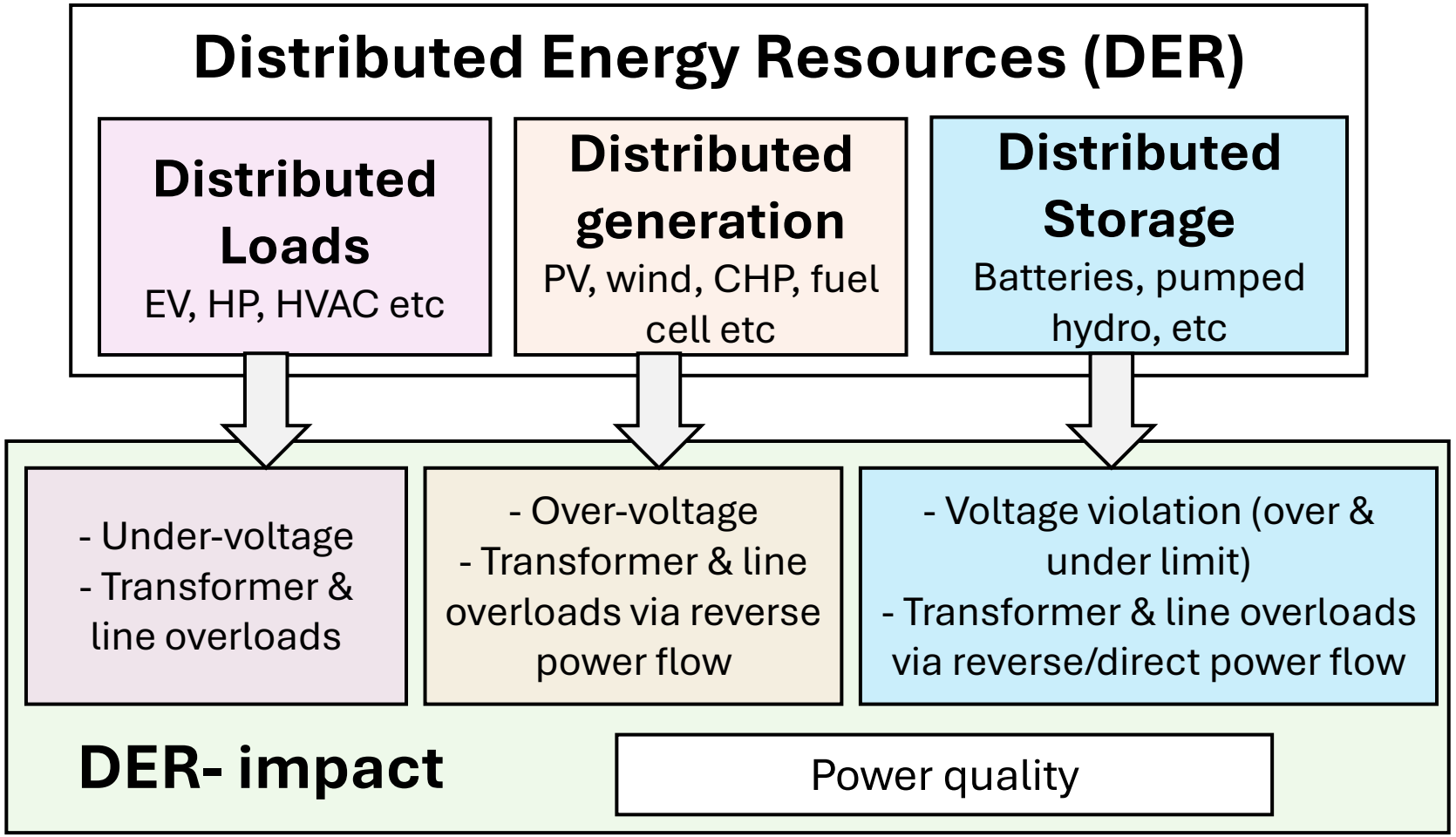
Source: https://www.gridx.ai/knowledge/distributed-energy-resources

What changes with distributed generation?



Courtesy: Wikipedia

Impact of DER connections on network



[Ref] Koirala, Arpan, Md Umar Hashmi, Reinhilde D'hulst, and Dirk Van Hertem. "Decoupled probabilistic feeder hosting capacity calculations using general polynomial chaos." *Electric Power Systems Research* 211 (2022): 108535.
[Ref] Fani, Hossein, Md Umar Hashmi, and Geert Deconinck. "Impact of electric vehicle charging simultaneity factor on the hosting capacity of LV feeder." *Sustainable Energy, Grids and Networks* 40 (2024): 101581.

Motivation

Behavioral changes in load consumption

New loads: EV charging & Heat pump load

- The sales have increased by **+50%** in 2022, compared to 2021 for the EVs
- The sales for heat pumps have also increased by **+60%** in 2022 compared to 2021

- High peaks
- Highly simultaneous

Quick introduction to power system flexibility

Motivation

Favorable policies

- §14a EnWG (Energiewirtschaftsgesetz) in Germany
  - makes it possible for grid operators to temporarily reduce the output of (*but not turn off*) controllable energy-consuming assets in the event of potential overloads
    - This rule applies to DERs with more than 4.2 kW load (guaranteed minimum power consumption)
  - DSO no longer be allowed to refuse or postpone the connection of these new assets due to insufficient grid capacity
  - *Fairness consideration*: The focus should not always be on "*dimming*" the same household just because the control would be most effective there.

- FERC Order 2222 in the USA
  - Enables DER participation in the electricity markets run by regional ISO/RTO through aggregation

- Fluvius local flexibility market is live as of Dec 2024
  - Fluvius is procuring both short and long-term flexibility products for active & reactive power→ acquired through the NODESTM activation market, known as ShortFlexTM, LongFlexTM, MaxUsage.
  - Entities participating in this initiative will be known as Flexibility Service Providers (FSPs).

Source:
[1] https://www.gesetze-im-internet.de/enwg_2005/__14a.html
[2] https://www.ferc.gov/ferc-order-no-2222-explainer-facilitating-participation-electricity-markets-distributed-energy
[3] https://nodesmarket.com/fluvius-market/
[4] https://www.gridx.ai/knowledge/paragraph-14a-enwg

What is flexibility (in power system) ?

- The technical ability of a power system unit to **modulate its power** feed-in or power out-feed over time
  - Active and reactive power (P and Q) modulation ↑ or ↓

- *ENTSO-e definition*: Power system flexibility is defined as the ability of a power system to effectively **manage the variability and uncertainty of electricity demand, supply, and grid availability** across a range of timeframes, from seconds to years.

- *Goal of flexibility*: Adjust to the variability of consumption and generation patterns and to grid availability
  - Via European Association for Storage of Energy (EASE)

- In practice: deviation from a Business-As-Usual situation (=baseline)
  - What is this baseline?

Importance of flexibility

- **Increasing share of DER** in power system
- Pressure on distribution network: frequency variation, congestion, forecast errors, … → **active elements**
- Flexibility facilitates more **efficient use of grid** without need for expensive network upgrades → becomes valuable economic asset
- For flexibility owners → **financial incentive** in terms of reduction in cost of energy consumption is prevalent

Flexibility parameters

Baseline for flexibility quantification

Flexibility quantification is often performed using a baseline.

With respect to baseline, the ramp up or down is measured

Historical Baseline: This baseline is calculated using historical load data, typically by taking the average load over a similar period in the past (e.g., the same day of the week, same time of day, and similar weather conditions).	$P_{\text{baseline}}(t) = \frac{1}{N}\sum_{i=1}^{N} P_{\text{historical},i}(t)$
Model-Based Baseline: This baseline is derived from a model that predicts the expected load based on various factors such as time of day, day of the week, weather conditions, and other relevant variables	$P_{\text{baseline}}(t) = f(t, \text{weather}, \text{other factors})$
Moving Average Baseline: This baseline is calculated using a moving average of the load over a recent time window.	$P_{\text{baseline}}(t) = \frac{1}{W}\sum_{i=t-W}^{t-1} P(i)$
Day-Ahead Baseline: This baseline is based on the day-ahead load forecast, which is typically provided by the utility or system operator.	$P_{\text{baseline}}(t) = P_{\text{forecast}}(t)$

$$\Delta P_{\text{up}}(t) = P_{\text{actual}}(t) - P_{\text{baseline}}(t)$$

$$\Delta P_{\text{down}}(t) = P_{\text{baseline}}(t) - P_{\text{actual}}(t)$$

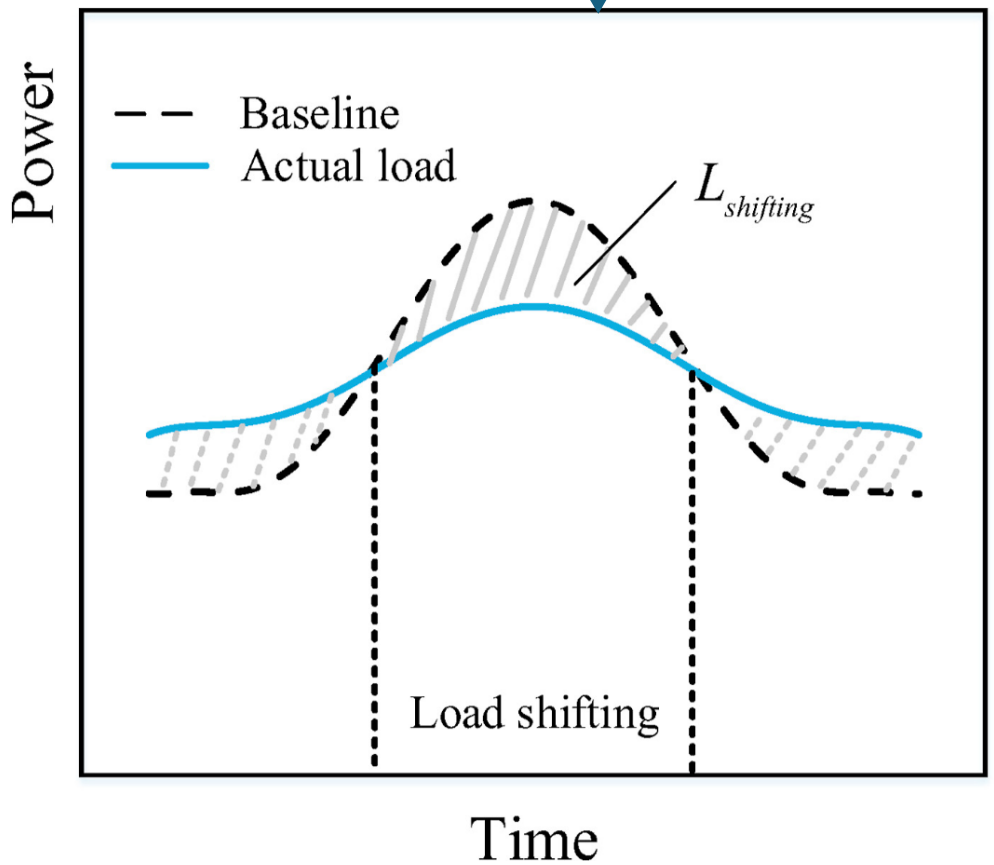


Figure source: Tang, Hong, and Shengwei Wang. "**Energy flexibility quantification of grid-responsive buildings: Energy flexibility index and assessment of their effectiveness for applications.**" *Energy* 221 (2021): 119756.

How to quantify flexibility?

What metrics to use?

- 5 metrics for flexibility quantification:
  - ➢ Active power capacity [kW]
  - ➢ Active power ramp rate [kW/min.]
  - ➢ Energy supply [kWh]
  - ➢ Reactive power capacity [kVAr]
  - ➢ Reactive power ramp rate [kVAr/min.]

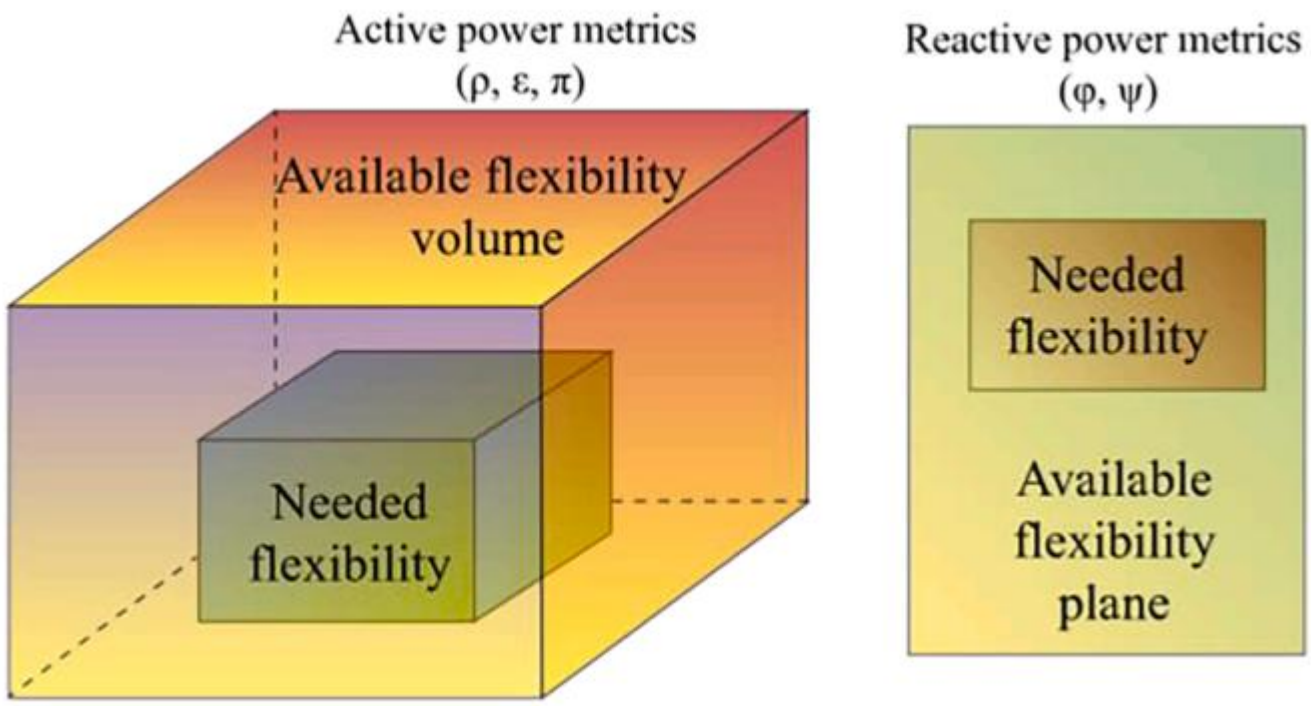


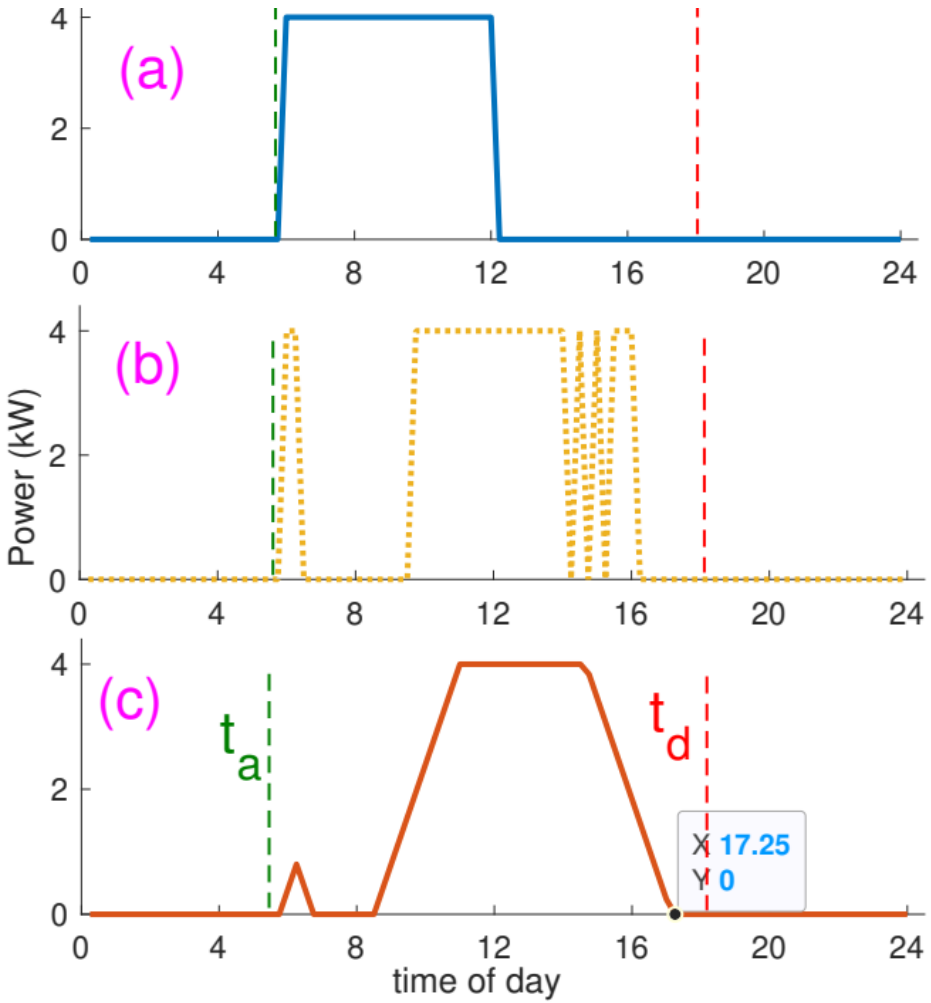
Figure source: Sarmiento-Vintimilla, Juan C., et al. "Assessment of the operational flexibility of virtual power plants to facilitate the integration of distributed energy resources and decision-making under uncertainty." *International Journal of Electrical Power & Energy Systems* 155 (2024): 109611.

Flexibility modelling

Is it important to model ramp rate? → Illustrative example

Flexibility model

- Deadline constraint
  - Total energy consumption target
- Ramping constraint
  - Device capability to consume or generate maximum power
- Ramp rate constraint
  - Delta P <= threshold



Charging profiles comparison:

(A) The nominal charging profile,

(B) the charging profile based on smart charging without ramp rate constraint,

(C) the charging profile for ramp rate constraint of is 10% of the ramp (kW) limit.

Case study: The electric vehicle is connected to the charger at 6 a.m. and leaves at 6 p.m. The connection time, thus, is 12 hours. However, the charging duration is only 6 hours and 15 minutes. The total charged energy is 25 kWh.

[Ref] Hashmi, M. U., Van Hertem, D., van der Meer, A., & Keane, A. (2024, October). **Linear energy storage and flexibility model with ramp rate, ramping, deadline and capacity constraints**. In *2024 IEEE PES Innovative Smart Grid Technologies Europe (ISGT EUROPE)* (pp. 1-6). IEEE.

Summary

- Flexibility is crucial for <u>economical viable network operation</u> with large scale integration of intermittent distributed energy resources

- **Heterogenous nature** of distribution network flexibility based (i) availability, (ii) management can be quantified for (a) power, (b) energy and (c) ramp rate capabilities

- With and without Baseline based **flexibility characterization is crucial**

Thank you for your attention!

Dr. Md Umar Hashmi
Research Expert, KU Leuven, Department of Electrical Engineering
Heverlee, 08.02.2023

mdumar.hashmi@kuleuven.be

https://github.com/umar-hashmi

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